

Carbon-plate insole use and self-reported injury prevalence among Italian competitive athletics athletes: a cross-sectional, Bayesian multilevel analysis of a two-phase recruited cohort

Anonymized Author 1

Anonymized Author 2

Abstract

Objective: To describe carbon-plate insole use during preparation and competition among competitive Italian athletics athletes, and to estimate the cross-sectional, period-matched association between reported carbon share and self-reported current-season injury.

Design: Cross-sectional survey with two-phase (voluntary, then targeted-recall) recruitment, poststratified to national age–sex population strata; reported in line with STROBE and the STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) extension.

Participants: FIDAL-affiliated athletics athletes (Under 20, Under 23, Senior categories) recruited nationally in 2026. Of 355 eligible, complete respondents, 352 (704 athlete-period records) formed the balanced analysis cohort.

Exposure: Carbon share, the proportion of reported weekly training-type frequency delivered in carbon-plated spikes across five training domains (long endurance, middle endurance, lactate, maximum velocity, technique), calculated separately for the preparation and competition periods.

Outcome: Self-reported injury occurring in the preparation period and/or the competition period of the current season.

Statistical methods: A Bayesian multilevel logistic model with period-specific intercepts, an athlete-level random intercept, a natural-spline exposure term among users, and adjustment for demographic, anthropometric, training-load, and discipline covariates. Estimands were posterior-standardized adjusted prevalences, differences, and ratios, reported with 95% credible intervals (CrI) and posterior probabilities. Poststratification weights aligned the sample to national age–sex population estimates.

Results: Carbon-plated spikes accounted for a mean 35.7% of reported preparation training and 47.7% of reported competition training, a stable pattern across sex and age strata. Among 193 injured athletes completing four adapted OSTRC-H2 impact items, the median severity score was 75 (IQR 58–92) and was highest for injuries spanning both periods (84, IQR 67–92). Adoption of carbon use (median user versus non-user) showed no clear association with injury in either period. Among competition-period users, moving from the 25th to 75th percentile of carbon share was associated with a 14.3 percentage-point higher adjusted injury prevalence (95% CrI 2.7 to 25.8; adjusted prevalence ratio 1.41, 95% CrI 1.06 to 1.86; posterior probability of a positive association 99.0%). The corresponding preparation-period dose contrast was unclear (–0.8 pp, 95% CrI –10.6 to 9.1).

Conclusion: Among competition users, a higher reported carbon share was associated with higher adjusted competition-period injury prevalence. Adoption itself and preparation-period dose were not clearly associated with injury. The cross-sectional, retrospective design does not support a causal interpretation.

Keywords: athletics; track and field; carbon-fiber plate; advanced footwear technology; injury surveillance; Bayesian multilevel model; STROBE-SIIS.

1 Introduction

Athletics (track and field) carries a substantial burden of health problems at the competitive tiers of participation [McKay et al., 2022]. A systematic review and meta-analysis of 216 studies in Olympic athletics disciplines reported a mean injured-athlete prevalence of 50.4% (range 2.4%–93.6%) and a mean incidence of 135.6 injuries per 1000 registered athletes, with injury patterns differing markedly by discipline group, tissue type, and period of the season [Edouard et al., 2020]. Roughly half of injuries in youth athletics academies were reported in the winter preparation period [Martínez-Silván et al., 2021], motivating separate description of preparation- and competition-period injury rather than a single season-level estimate.

How injury is registered materially changes the burden observed. Surveillance restricted to absence, complete sports cessation, or medical attention can under-ascertain overuse problems because athletes may continue training and competing despite symptoms and impaired function [Clarsen et al., 2013]. The Oslo Sports Trauma Research Center Questionnaire on Health Problems (OSTRC-H) therefore complements time-loss registration with direct athlete reports across four impact domains: sports participation, training volume, performance, and symptoms [Clarsen et al., 2014, 2020]. These domains provide an athlete-reported measure of sport-specific health burden; they should not be interpreted as a generic quality-of-life scale.

Since 2020–2021, competitive athletics has undergone a rapid equipment transition with the introduction of “advanced footwear technology” (AFT) spikes: a rigid, longitudinally curved carbon-fiber plate embedded in a high-resilience foam midsole, first validated in road racing (Nike Vaporfly, Breaking2 project) and subsequently extended to track spikes for sprint and middle-distance events [Healey et al., 2022]. The stated biomechanical rationale is that plate-driven longitudinal bending stiffness reduces energy dissipation at the metatarsophalangeal joints and shifts the center of pressure anteriorly, increasing horizontal propulsive force [Willwacher et al., 2016]. Comparative analyses of world-leading sprint performance before and after the introduction of AFT spikes have documented step changes in top-100 performance lists coincident with the technology’s diffusion [Mason et al., 2023].

The same mechanical changes that plausibly improve performance also alter loading at the ankle and forefoot. A recent kinetic comparison of traditional and carbon-plated sprint spikes in elite sprinters found materially higher ankle joint contact force and higher peak plantarflexor muscle force in the carbon-plated condition, with pronounced inter-individual variability [Gerlach et al., 2025]. Case reports and narrative reviews have raised the possibility of increased Achilles tendon load and of altered forces at the navicular and cuneiform bones, both recognized high-risk sites for bone stress injury in runners [Tenforde et al., 2023]. Evidence on ground-reaction-force-mediated risk is mixed, and no consensus exists on whether increased longitudinal bending stiffness is protective, neutral, or hazardous once inter-individual differences in plantarflexor strength are taken into account [Willwacher et al., 2016, Gerlach et al., 2025].

Despite this plausible mechanistic pathway, direct epidemiological evidence on carbon-plate insole use and injury in competitive athletics is essentially absent: available data are limited to isolated case reports and small mechanistic cohorts, none of which were designed to estimate population-level injury associations [Tenforde et al., 2023, Gerlach et al., 2025]. Established risk factors for running-related injury more broadly include age, body mass index, weekly training frequency, footwear, years of sporting experience, and prior injury [Taunton et al., 2003], all of which needed to be accounted for before any exposure–outcome association attributable to carbon-plate use could be considered informative.

This study was designed to close part of this gap by describing carbon-plate insole use among a national sample of Italian competitive athletics athletes and by estimating the cross-sectional,

period-matched association between reported carbon share and self-reported injury during the preparation and competition periods of a single season. Consistent with the Injury Definitions Concept Framework and the athletics-specific consensus statement on injury and illness surveillance [Timpka et al., 2014, 2015], injury was defined broadly as a physical complaint or observable tissue damage attributable to the transfer of energy experienced by an athlete during training or competition participation in athletics, irrespective of whether medical attention was sought or of the consequences for participation [Bahr et al.]. The primary hypothesis, formulated a priori, was that greater reported carbon share would be associated with higher adjusted injury prevalence; a secondary, exploratory aim was to describe whether carbon-plate adoption and share differed by age group and sex, and whether the exposure–outcome association was stable under population reweighting and alternative dose–response specifications.

2 Methods

2.1 Study design and reporting

This is a retrospective, cross-sectional survey study of competitive athletics athletes affiliated with the Federazione Italiana di Atletica Leggera (FIDAL). Reporting follows the STROBE statement and its athletics-relevant STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) extension [von Elm et al., 2007, Bahr et al.]; the completed checklist, with the manuscript line number supporting each item, is given in Appendix A.

2.2 Study registration

This study was not registered in a public trial or study registry (for example, ClinicalTrials.gov or an equivalent observational-study registry). It was, however, prospectively specified through a complete study protocol, submitted in Italian and reviewed and approved by the Comitato Etico of the University of Turin (protocol n. 0453001, approved 23 June 2026; Section 2.3) before survey deployment. The protocol specified the study population, the two-phase recruitment and poststratification strategy, the exposure and outcome definitions, and the primary hypothesis (Section 1) in advance of data collection and of any outcome analysis. This ethics-committee-reviewed protocol is the prospective record of the study design; it is not, however, a substitute for public registration, and the absence of the latter is noted as a limit on independent, external verification of pre-specification (Section 4.4).

2.3 Setting and ethics

Data collection used a structured, self-administered online survey covering demographic, anthropometric, training, footwear, and injury-history items for the current competitive season, separately for the preparation and competition periods. Study data were collected and managed using REDCap electronic data capture tools hosted at the Department of Clinical and Biological Sciences, University of Turin [Harris et al., 2009, 2019]. REDCap (Research Electronic Data Capture) is a secure, web-based software platform designed to support data capture for research studies, providing (1) an intuitive interface for validated data capture, (2) audit trails for tracking data manipulation and export procedures, (3) automated export procedures for seamless data downloads to common statistical packages, and (4) procedures for data integration and interoperability with external sources.

The study was approved by the Comitato Etico of the University of Turin (protocol n. 0453001, approved 23 June 2026). Participation required electronic informed consent; no financial incentive

was offered. Individual survey responses were stored under a study identifier disconnected from athlete names at the point of analysis; only aggregate, de-identified data and model outputs are reported here, in line with the data-privacy provisions of the Declaration of Helsinki.

2.4 Participants and two-phase recruitment

Eligible athletes were FIDAL-affiliated competitive athletes in the Under 20, Under 23, or Senior (*assoluta*) categories, across seven athletics disciplines with adequate cell support in the sample (sprint, middle distance, long distance, hurdles, horizontal jumps, pole vault, combined events; four further disciplines had zero respondents and were dropped from discipline-level adjustment; Appendix Table 14).

Recruitment proceeded in two phases to reduce the risk that a purely voluntary sample would over-represent already engaged or higher-exposure athletes. In Phase 1, the survey was distributed openly through FIDAL-affiliated clubs and athlete networks; enrollment was entirely voluntary. National population totals by age category, sex, and discipline group were estimated from FIDAL competition-registry history (Appendix Table 15). Study size was planned for precision of a population proportion rather than power for a specific exposure association. For each candidate sample size n , the worst-case count $x = \lfloor n/2 \rfloor$ was combined with a Beta(1,1) prior, giving a Beta($1+x$, $1+n-x$) posterior; the smallest n whose equal-tailed 95% credible interval had half-width no greater than 0.06 was selected. This rule gave $n = 264$ from the estimated eligible frame of 1328 athletes (19.9%); targets were allocated proportionally across the 42 age-by-sex-by-discipline strata using largest-remainder rounding (Appendix Table 16). The calculation established a minimum precision and population-coverage target, not representativeness by itself. After Phase 1 closed, strata that had not reached their target count were identified, and Phase 2 undertook targeted recall–direct outreach to athletes and clubs within under-represented age-by-sex-by-discipline strata–until every stratum met or approached its target or the recruitment window closed. Detailed recruitment-monitoring cells are retained privately to avoid disclosing small-cell participation patterns. This two-phase design does not produce a probability sample with known, non-zero, equal selection probabilities in every stratum; population coverage was supported by targeted recruitment, and residual over- and under-representation was corrected analytically (Section 2.9) rather than assumed away.

Of all survey respondents, 355 provided consent, met calculated eligibility, and completed the survey (`survey_atleta_it_complete = 2`). One athlete was excluded for incomplete injury-timing data, and two further athletes were excluded because a period (preparation or competition) had a zero total eligible training-frequency score, precluding a carbon-share denominator. The final balanced analysis cohort comprised 352 athletes contributing 704 athlete-period records (one preparation and one competition record per athlete; Table 3).

2.5 Exposure: carbon-plate insole use

For each period (preparation, competition) and each of five training domains—long endurance, middle endurance, lactate, maximum velocity, and technique—athletes reported an ordinal weekly training-type frequency (0, 1, 2, 3, 4, or 5-or-more sessions per week, with 5-or-more coded as 5) separately for sessions performed in carbon-plated spikes and sessions performed without them. *Carbon share* for a given period was defined as

$$\text{carbon share} = \frac{\sum \text{carbon-frequency items}}{\sum \text{carbon-frequency items} + \sum \text{non-carbon-frequency items}},$$

summed across the five domains. Gym/strength conditioning (*palestra*) has no carbon-eligible equivalent and was retained only as a non-carbon adjustment covariate, not in the carbon-share denom-

inator. Carbon share is therefore a composition of reported ordinal training-type frequency, not a directly measured training-time proportion or shoe-wear duration; this distinction is carried through to the interpretation of all exposure estimands. Exposure was decomposed into two components analyzed jointly: *adoption* (any carbon use versus none, `carbon_frequency_sum > 0`) and, among users, continuous *dose* (carbon share).

2.6 Outcome: period-specific injury

The outcome was self-reported injury attributable to athletics training or competition participation during the current season, following the athletics-specific consensus definition [Bahr et al., Timpka et al., 2014]: any physical complaint or observable tissue damage resulting from the transfer of energy experienced by an athlete during participation, regardless of medical attention sought or time-loss consequence. Athletes reported the number of current-season injuries (0, 1, or 2) and, for each, whether onset occurred in the preparation period, the competition period, or both (coded 1, 2, or 3 respectively). Preparation-period injury was defined as any reported injury coded 1 or 3; competition-period injury as any reported injury coded 2 or 3. Athletes could therefore contribute a positive outcome to one or both periods.

For each reported injury, athletes also selected the anatomical location (hip, groin, thigh, lower leg, ankle, foot, or head/upper limb/trunk; multi-select) and, only for injuries localized to the lower limb, the injury type (joint sprain, tendon rupture, ligament injury, muscle strain, bone stress injury or stress fracture, tendinopathy, plantar fasciopathy, or other; multi-select, adapted from consensus-recommended categories [Bahr et al.]). These location and type items were not used as covariates or outcomes in the primary Bayesian model; they are reported descriptively (Section 3.5) to characterize the injury pattern in this cohort, following STROBE-SIIS item SIIS17.1. Injury severity (time-loss duration) was also collected but is outside the scope of the present analysis.

Athletes reporting at least one current-season injury were also presented with four items adapted from the updated OSTRC-H2, covering the injury’s impact on sports participation, training volume, sports performance, and symptoms [Clarsen et al., 2014, 2020]. The four ordered responses were scored 0, 8, 17, and 25, then summed to an adapted OSTRC severity score from 0 (no reported impact) to 100 (maximum reported impact). Because these items were administered once retrospectively and only after a current-season injury was reported, the score is used solely as a descriptive athlete-reported impact measure; it is not a prospective weekly OSTRC prevalence or cumulative-burden estimate.

2.7 Covariates

Adjustment covariates were age (years), competition sex, height (cm), body mass (kg), years of athletics experience, injury in either of the two preceding seasons (binary), period-matched total training-type-frequency scores for each of the five domains and for *palestra*, and indicator variables for the seven discipline categories with non-zero support. These variables were selected a priori based on previously reported risk factors for running-related injury [Taunton et al., 2003] and on the requirement, specified in the STROBE-SIIS extension, to account for how athletes were categorized by sport, discipline, and competitive level [Bahr et al.]. Continuous covariates were standardized (zero mean, unit variance) before model fitting.

2.8 Statistical methods

All inference is Bayesian [Gelman et al., 2013]; no null-hypothesis-testing framework or *p*-values are reported. Results are summarized as posterior means, medians, standard deviations, and 95%

credible intervals (CrI, equal-tailed 2.5th–97.5th percentiles), together with posterior probabilities that an association exceeds zero (or one, for ratios), exceeds a pre-specified practically-relevant threshold (5 percentage points for differences, a ratio of 1.10), or falls within a region of practical equivalence to no association (± 2 percentage points; ratio 0.95–1.05). Models were fit in Stan [Carpenter et al., 2017] via its R interface.

The adapted OSTRC scores were summarized by mutually exclusive injury-period groups (preparation only, competition only, or both periods) using medians and interquartile ranges. As a descriptive effect-size aid, standardized mean differences (SMD) compared preparation-only and both-period groups with the competition-only group, calculated as the mean-score difference divided by the square root of the average of the two group variances. No injured-versus-uninjured OSTRC comparison was made because the items were not administered to uninjured athletes.

2.8.1 Primary model

Each athlete contributed one Bernoulli outcome per period. A shared athlete-level random intercept linked the two outcomes, following a cluster-specific logistic formulation for correlated binary data [Neuhaus et al., 1991]. For athlete i in period p ,

$$Y_{ip} \sim \text{Bernoulli}(\pi_{ip}), \quad \text{logit}(\pi_{ip}) = \alpha_p + c_{ip} + a_{ip} + \delta_i + u_i, \quad u_i \sim \mathcal{N}(0, \sigma_u^2).$$

Here c_{ip} is the carbon-use contribution:

$$c_{ip} = \begin{cases} 0, & \text{for a non-user,} \\ \beta_p^{\text{any}} + f_p(\text{share}_{ip}), & \text{for a carbon user.} \end{cases}$$

The remaining terms have direct roles: α_p is the period-specific intercept, a_{ip} combines the athlete-level and period-matched covariates of Section 2.7, δ_i is the partially pooled discipline contribution, and u_i is the athlete random intercept. The function f_p is a period-specific three-degree-of-freedom natural spline, fitted among users with a common basis and centered at the pooled median user share (0.50); its internal knots were 0.375 and 0.571, with boundary knots 0 and 1. Consequently, β_p^{any} compares a median-share user with a non-user on the conditional log-odds scale, while f_p describes dose among users. The athlete and discipline terms were parameterized non-centered, with discipline coefficients partially pooled through a shared scale τ_{disc} . Priors were regularizing and weakly informative on the standardized log-odds scale: $\alpha_p \sim \mathcal{N}(\text{logit}(0.4), 1.5^2)$, exposure coefficients $\mathcal{N}(0, 0.7^2)$, covariate coefficients $\mathcal{N}(0, 0.5^2)$, $\tau_{\text{disc}} \sim \text{Half-N}(0, 0.5^2)$, and $\sigma_u \sim \text{Half-N}(0, 1^2)$. A matched baseline model retained the identical outcome, covariate, discipline, and athlete-random-intercept structure but removed both carbon-use terms, providing a no-exposure comparator for predictive-validity assessment (Section 2.8.3).

Figure 1 summarizes the primary model’s observed inputs, parameter hierarchy, and repeated athlete-period outcome structure.

2.8.2 Estimands and marginal standardization

Posterior draws were used to compute marginal (population-standardized) adjusted prevalences by averaging model predictions over the observed empirical distribution of covariates and discipline indicators within each period, and by integrating the athlete random intercept using 15-point Gauss–Hermite quadrature rather than conditioning on a fixed value of u_i . For each period, adjusted prevalence was computed at four exposure scenarios: non-user, and user at the 25th (low), 50th (median), and 75th (high) percentile of the observed among-user share distribution for that

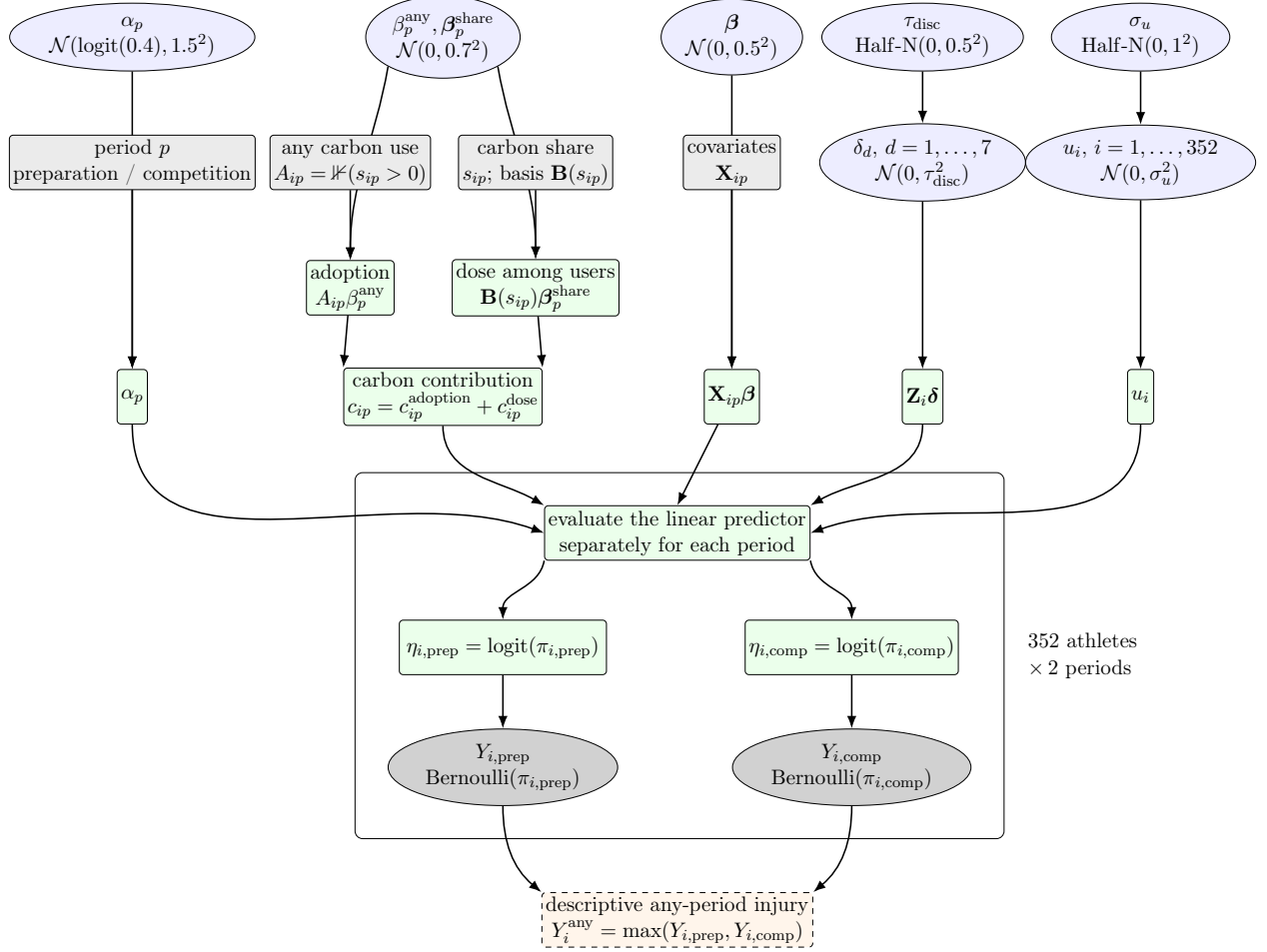


Figure 1: Graphical representation of the primary Bayesian multilevel model. Grey boxes are observed inputs, blue ellipses are estimated parameters, green boxes are deterministic contributions to the linear predictor, and shaded ellipses are the two observed period-specific injury outcomes, jointly enclosed in the solid plate (352 athletes $\times$ 2 periods). The carbon contribution c_{ip} combines adoption (A_{ip} , any carbon use) and dose ($\mathbf{B}(s_{ip})$, the centered three-degree-of-freedom natural-spline basis for carbon share among users). $\mathbf{Z}_i$ denotes the seven discipline indicators underlying the partially pooled coefficients δ_d , and u_i the shared athlete random intercept. The common model structure is evaluated separately as $\eta_{i,\text{prep}}$ and $\eta_{i,\text{comp}}$; each predictor points only to its corresponding outcome. The dashed box is the derived any-period injury indicator reported descriptively, outside the probabilistic-model plate, and is not an additional likelihood term.

period. Two primary contrasts were derived: the *adoption* contrast (median user vs. non-user; prevalence difference and ratio) and the *dose* contrast (75th vs. 25th percentile among users; prevalence difference and ratio).

2.8.3 Validation and sensitivity analyses

Predictive comparison between the primary (exposure) and baseline (no-exposure) models used Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) [Vehtari et al., 2017], computed on the athlete-grouped joint log-likelihood (both period records per athlete evaluated jointly, with the athlete random intercept marginalized by Gauss–Hermite quadrature before leave-one-athlete-out prediction), reported as the expected log predictive density (ELPD) difference with its standard error. Posterior predictive checks compared observed and model-replicated counts of injured athletes per period. Two pre-specified sensitivity analyses tested structural assumptions of the primary specification: (1) a linear carbon-share term in place of the natural spline; (2) wider priors (exposure coefficients $\mathcal{N}(0, 1.2^2)$, covariate coefficients $\mathcal{N}(0, 1.0^2)$). Sensitivity models were compared with the primary model as robustness checks on the same estimands, not as competing substantive hypotheses.

2.9 Population reweighting

Because the two-phase recruitment (Section 2.4) could not guarantee equal selection probabilities across age-by-sex strata, a poststratification step [Gelman, 2007] reweighted the analysis sample to FIDAL national population estimates for six age-by-sex strata (Under 20, Under 23, Senior $\times$ Men, Women; Appendix Table 17). Stratum model weights were the ratio of the population share to the observed sample share, further regularized to avoid extreme weights. The Kish effective sample size after weighting was 340.5 of 352 (97%), indicating a modest efficiency loss from reweighting. Weighted estimands were computed by combining the athlete-level posterior predictions of Section 2.8.2 with model weights before marginalization, and are reported alongside the unweighted (sample-representative) estimands as a sensitivity check on generalizability to the national competitive population.

2.10 Subgroup analyses

As a pre-specified descriptive and exploratory extension, the primary adjusted estimands were re-computed conditional on sex (men, women) and on age group (Under 20, Under 23, Senior), holding the same model structure and covariate adjustment, to assess whether the direction and approximate magnitude of the competition-period dose association were consistent across these strata. These subgroup estimates are not powered as confirmatory tests of effect-modification and are reported as descriptive, hypothesis-generating comparisons.

2.11 Sources of bias

The survey instrument underwent content-validity and face-validity review by athletics-medicine content experts and prospective respondents before deployment, but no formal test-retest reliability or criterion-validity assessment against objectively measured training data (for example, coach-logged sessions or shoe-embedded wear sensors) was conducted; this is treated as a limitation rather than a validated measurement property (Section 4.4). Both exposure and outcome were self-reported for the same recall period by the same respondent, which admits recall bias and the possibility that athletes’ beliefs about carbon-plate risk influenced how they reported either quantity. Reporting

was not restricted to medically trained staff; athletes reported their own injuries and training without independent clinical verification, following the broad, non-medical-attention-dependent injury definition of Section 2.6, which was chosen specifically to reduce under-ascertainment of injuries that did not prompt a clinical visit at the cost of relying on athlete-reported symptom recognition rather than diagnosis.

2.12 Missing data and other analyses

The analysis used a complete-case cohort by design: eligibility required a fully completed survey, and the three further exclusions in Section 2.4 addressed incomplete injury timing or a structurally undefined carbon-share denominator. No further imputation was performed.

2.13 Software

The fitted models used a Bernoulli likelihood with a logit link. Posterior sampling used Stan’s No-U-Turn Sampler (NUTS), a Hamiltonian Monte Carlo algorithm, through the `rstan` interface. For each model, four independent chains each ran for 2000 iterations; the first 1000 were used for warm-up and adaptation and discarded, leaving 1000 retained draws per chain and 4000 retained draws in total to approximate the posterior distribution. The target acceptance probability was set to `adapt_delta`=0.98 and the maximum tree depth to 12. Runs used base seed 20260719 with deterministic model-specific offsets (100, 200, 300, and 400 for the primary association, baseline, linear-sensitivity, and wide-prior models, respectively; 500 for the population-weighted model). Analyses used the `loo` package for PSIS-LOO cross-validation and the base `splines` package for natural splines. Full software versions are reported in Appendix B.

3 Results

3.1 Sample flow and recruitment

Table 3 summarizes the flow from survey completion to the balanced analysis cohort of 352 athletes and 704 athlete-period records. Phase 2 targeted recall concentrated additional enrollment in Under 20 and Under 23 strata, particularly hurdles, horizontal jumps, and combined events, which the voluntary Phase 1 sample under-represented relative to the FIDAL population estimate. After poststratification (Section 2.9; Appendix Table 17), the Kish effective sample size was 340.5 of 352 analyzed athletes (97%), indicating that reweighting did not substantially inflate posterior uncertainty.

3.2 Baseline characteristics

Table 2 compares baseline characteristics by competition sex, the principal descriptive contrast for this cohort, alongside the corresponding population-weighted standardized mean differences (SMD). As expected for a sex comparison, height and weight showed large SMDs (−1.81 and −1.83); demographic, injury-history, carbon-exposure, and discipline-distribution SMDs were mostly below the conventional 0.20 threshold for a small imbalance [Hayes-Larson et al., 2019], with horizontal jumps and combined events showing modest over-representation of women (SMD 0.22–0.23). Reweighting to national population strata (weighted SMD column) did not materially change this pattern, supporting the sample as broadly representative of the target competitive population once poststratified.

3.3 Descriptive carbon-plate insole use

Across the balanced cohort, carbon-plated spikes accounted for a mean 35.7% of reported preparation-period training-type frequency and 47.7% of reported competition-period training-type frequency (Table 4). This pattern—roughly two-fifths of preparation training and roughly half of competition training reported in carbon-plated spikes—was stable across sex and age-group strata (Table 4, Figure 4), with mean shares differing by at most 4 percentage points between men and women or across the three age groups in either period, and with the full distributions—not only their means—overlapping closely between strata. Adoption (any carbon use) was high in both periods: 81.5% of athletes reported any preparation carbon use and 87.5% reported any competition carbon use (Table 2), consistent with carbon-plated spikes functioning as the prevailing rather than niche equipment choice in this population.

3.4 Specialty-group-stratified characteristics

To summarize the seven disciplines with limited individual cell sizes (19–107 athletes; Appendix Table 14), they were collapsed into four broader, physiologically motivated specialty groups—sprint/hurdles, endurance (middle and long distance), jumps (horizontal jumps or pole vault), and combined events. Tables 5 and 6 report baseline characteristics, injury prevalence, and carbon-plate use by group. Because disciplines and, consequently, groups are multi-select, 26 of 352 athletes belong to more than one group. Two patterns stand out as original descriptive findings of this cohort.

First, competition-period carbon share tracked the explosive-versus-endurance character of the group rather than being uniform across athletics. The sprint/hurdles group (the largest, $n = 169$) had the highest competition carbon share (0.56) and the largest preparation-to-competition rise (+0.17), while the endurance group ($n = 109$) had the lowest carbon share in both periods (0.31 and 0.34). This is consistent with the biomechanical rationale in Section 1: carbon-plate stiffness benefits are most established for short-duration, high-power efforts, and this cohort’s self-reported equipment allocation mirrors that distinction even though the survey did not ask athletes about performance rationale directly.

Second, injury prevalence did not track carbon share in the same way. The jumps group ($n = 62$) stood out with the highest prior-injury history (83.9%) and the highest competition-period injury prevalence (51.6%) of the four groups, despite an intermediate carbon share (0.47)—higher than endurance but lower than sprint/hurdles—indicating that the elevated injury burden in jumping disciplines is unlikely to be explained by carbon-plate dose alone and more plausibly reflects the landing and eccentric loading demands specific to jumping events; the primary Bayesian model (Section 3.7) already adjusts for discipline using partially pooled indicators specifically to avoid attributing this discipline-driven variation to carbon exposure. The combined-events group ($n = 41$) showed a distinct hybrid profile: a carbon share close to sprint/hurdles (0.52) paired with an endurance-like training volume (preparation and competition medians of 9) and an elevated any-injury prevalence (70.7%), consistent with the mixed physiological demands of multi-event competition and with the higher, more heterogeneous musculoskeletal load noted for plyometric and multi-skill disciplines in prior athletics injury literature [Edouard et al., 2020].

3.5 Injury location and type

The 204 injured athletes reported 259 injuries in total (149 athletes with one injury, 55 with two), a median of 1 injury per affected athlete. Table 7 reports the anatomical location of these injuries and Figure 2 displays both the location and type distributions together. The thigh was, by a substantial margin, the single most commonly reported location (97 of 259 injuries, 37.5%; affecting 80 of 204

injured athletes, 39.2%), followed by the lower leg (25.1% of injuries) and the foot (19.3%); together, the hip, groin, thigh, lower leg, ankle, and foot accounted for 95.8% of all reported injuries, with only 4.2% localized to the head, upper limb, or trunk.

For the 248 lower-limb injuries for which an injury type was recorded (Table 8, Figure 2), tendinopathy (28.6%) and bone stress injury or stress fracture (26.2%) were the two most frequently reported types, together accounting for more than half of classified lower-limb injuries and each individually more common than muscle strain (22.2%). This is a distinctive feature of this cohort relative to general athletics injury surveillance, where muscle injury is typically the most frequently reported tissue type by a wide margin and tendon and bone injuries are reported considerably less often [Edouard et al., 2020].

Cross-tabulating type against location (Table 9) shows that these two leading types were not distributed the same way anatomically. Tendinopathy was disproportionately distal relative to the overall injury-location pattern: foot (35.2% of the 71 tendinopathy injuries) and lower leg (33.8%) were more prominent locations for tendinopathy than for the injury sample as a whole (19.3% and 25.1% respectively; Table 7), and ankle involvement (22.5%) was likewise higher than the sample-wide 13.9%—a pattern compatible with Achilles- and plantar-tendon structures at the distal lower limb, consistent with the biomechanical literature on carbon-plate loading of the ankle and forefoot [Gerlach et al., 2025]. Bone stress injury or stress fracture, in contrast, was thigh-dominant (58.5% of the 65 bone stress injuries), a pattern closely tracking the overall injury-location distribution and similar to muscle strain (74.5% thigh) rather than being disproportionately distal; foot involvement among bone stress injuries (15.4%) was, if anything, lower than the sample-wide rate. The elevated overall frequency of bone stress injury in this cohort is therefore a genuine descriptive finding, but this cohort’s data do not show the foot/navicular-specific concentration that would most directly support the mechanistic link to carbon-plate loading proposed for that injury type [Tenforde et al., 2023]; the survey’s type item does not further subclassify bone site (for example, navicular versus tibial versus metatarsal), so this distinction cannot be resolved further with the present data. Taken together, the location-by-type pattern strengthens the tendinopathy–carbon-plate hypothesis specifically, while leaving the bone-stress-injury hypothesis only partially supported. This is reported as a hypothesis-generating observation: the present cross-sectional design and sample size do not support a formal type-stratified test of the carbon-share association, and confirming a tendinopathy-specific association with carbon-plate dose, or resolving the anatomical site of the bone stress injuries observed here, would require a dedicated, adequately powered follow-up.

3.6 Athlete-reported injury burden

Of the 204 injured athletes, 193 (94.6%) completed all four adapted OSTRC-H2 impact items. The median total severity score was 75 (IQR 58–92) overall (Table 1). Scores were similar for athletes injured only during preparation (median 71.5, IQR 50–90) and only during competition (67, IQR 58–84; SMD -0.07), while athletes reporting injury in both periods had a higher median score of 84 (IQR 67–92; SMD 0.41 versus competition only). The largest component difference was for symptoms in the both-period group (SMD 0.51). These results describe impact among athletes already classified as injured; they are not a second injury-prevalence analysis.

3.7 Primary adjusted associations

Figure 3 shows the observed distribution of carbon share by period; Figure 7 displays the fitted among-user spline, and Figure 6 and Table 10 summarize the primary adjusted estimands. Adjusted preparation-period injury prevalence was 30.1% (95% CrI 21.0–40.2%) for non-users and 35.5% (95%

Table 1: Adapted OSTRC-H2 impact scores among 193 injured athletes with complete responses, stratified by mutually exclusive injury period. Values are median [Q1, Q3]. Each component ranges from 0 to 25 and total severity from 0 to 100; higher scores indicate greater impact. SMDs compare group means with the competition-only group and are descriptive.

Domain	Overall (<i>n</i> = 193)	Preparation only (<i>n</i> = 50)	Competition only (<i>n</i> = 84)	Both periods (<i>n</i> = 59)	SMD vs competition	
					Preparation	Both
Participation	17 [17, 25]	17 [17, 25]	17 [8, 25]	17 [17, 25]	0.14	0.19
Training volume	17 [17, 25]	17 [8, 25]	17 [8, 25]	25 [17, 25]	0.01	0.29
Performance	17 [17, 25]	17 [8, 25]	17 [17, 25]	25 [17, 25]	−0.35	0.22
Symptoms	17 [8, 25]	17 [8, 17]	17 [8, 17]	17 [17, 25]	−0.01	0.51
Total severity	75 [58, 92]	71.5 [50, 90]	67 [58, 84]	84 [67, 92]	−0.07	0.41

Injury location and type in the balanced cohort

Multi-select items; percentages sum to more than 100% within each panel

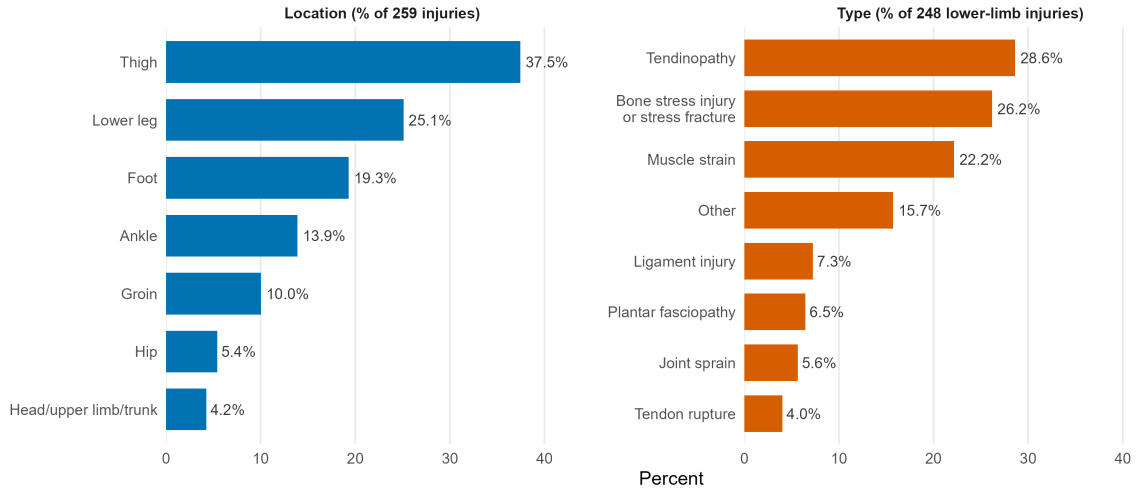


Figure 2: Anatomical location (% of 259 injuries) and type (% of 248 lower-limb injuries with a recorded type) of injuries reported by the balanced analysis cohort. Both items are multi-select, so percentages within each panel sum to more than 100%.

CrI 29.6–41.7%) for median-share users; the adoption difference of 5.4 percentage points (pp) (95% CrI –6.3 to 16.2) had a posterior probability of 82.0% of being positive, well short of the threshold used here to describe an association as clear. The preparation-period dose contrast (75th vs. 25th percentile share among users) was –0.8 pp (95% CrI –10.6 to 9.1; posterior probability of a positive association 43.8%), that is, centered near no association.

Adjusted competition-period injury prevalence was 38.1% (95% CrI 26.6–50.4%) for non-users and 43.0% (95% CrI 36.5–49.4%) for median-share users, an adoption difference of 4.8 pp (95% CrI –8.7 to 17.7; posterior probability 77.1%). In contrast, among competition-period users, moving from the 25th percentile (36.0% share) to the 75th percentile (71.4% share) of carbon share was associated with adjusted prevalence rising from 36.0% (95% CrI 28.5–44.0%) to 50.3% (95% CrI 42.4–58.2%): a difference of 14.3 pp (95% CrI 2.7 to 25.8) and a ratio of 1.41 (95% CrI 1.06 to 1.86), with a posterior probability of 99.0% that the association was positive and 94.0% that it exceeded the pre-specified 5-pp threshold for a practically relevant association (Table 10). This competition-period dose association was the only one of the four primary contrasts for which the credible interval excluded zero.

3.8 Model validation

Both the primary (exposure) and matched baseline (no-exposure) models showed acceptable sampling behavior: maximum $\hat{R} \leq 1.003$, zero divergent transitions, zero maximum-tree-depth hits, and minimum E-BFMI above 0.66 in both models (Appendix Table 20). Posterior predictive checks reproduced the observed injury counts closely: the full posterior predictive distribution of replicated injured-athlete counts was centered on 115.5 (observed 115) for preparation and 149.5 (observed 149) for competition, with the observed count falling centrally within each replicated distribution (Figure 5). Every athlete-level PSIS-LOO Pareto- k value was at most 0.5 in both models, supporting the reliability of the leave-one-out estimates. The exposure model’s athlete-grouped ELPD exceeded the baseline model’s by 2.22 (SE 2.30) (Appendix Table 20): the exposure terms did not clearly improve held-out predictive performance, even though the competition-period dose contrast itself had a high posterior probability of being positive. These two findings are compatible: a single, spline-parameterized, period-specific exposure effect concentrated in a subset of the competition-period users can produce a credible marginal association without moving whole-model out-of-sample predictive accuracy by a detectable amount in a cohort of this size.

3.9 Sensitivity analyses

The direction of the competition-period dose association was stable across both pre-specified sensitivity analyses, but its magnitude was not (Table 11, Figure 8). Under a linear carbon-share parameterization, the association reduced to 6.0 pp (95% CrI –0.3 to 12.6; posterior probability 96.9%), with the credible interval marginally including zero. Under wider priors, the association increased to 18.7 pp (95% CrI 5.2 to 31.9; posterior probability 99.5%). Taken together, the consistent positive direction across three structurally different specifications supports treating the competition-period dose association as a genuine feature of these data, while the sensitivity of its magnitude to the assumed dose–response shape argues against reporting 14.3 pp as a stable, precise effect size.

3.10 Population-weighted and subgroup estimates

After poststratification to national age–sex population strata, the competition-period dose association changed little: 14.9 pp (95% CrI 3.3 to 26.4) weighted versus 14.3 pp (95% CrI 2.7 to 25.8) unweighted (Appendix Table 19, Figure 9, Figure 10). The two adoption contrasts and the

465 preparation-period dose contrast likewise remained centered near no association after weighting.
 466 Conditional on sex or age group, the competition-period dose association retained a consistent
 467 point estimate (14.3–15.5 pp) and an essentially unchanged posterior probability of a positive asso-
 468 ciation (99.4% in every stratum examined; Appendix Table 18), providing no descriptive evidence
 469 that the association was concentrated in, or specific to, a single sex or age-group stratum.

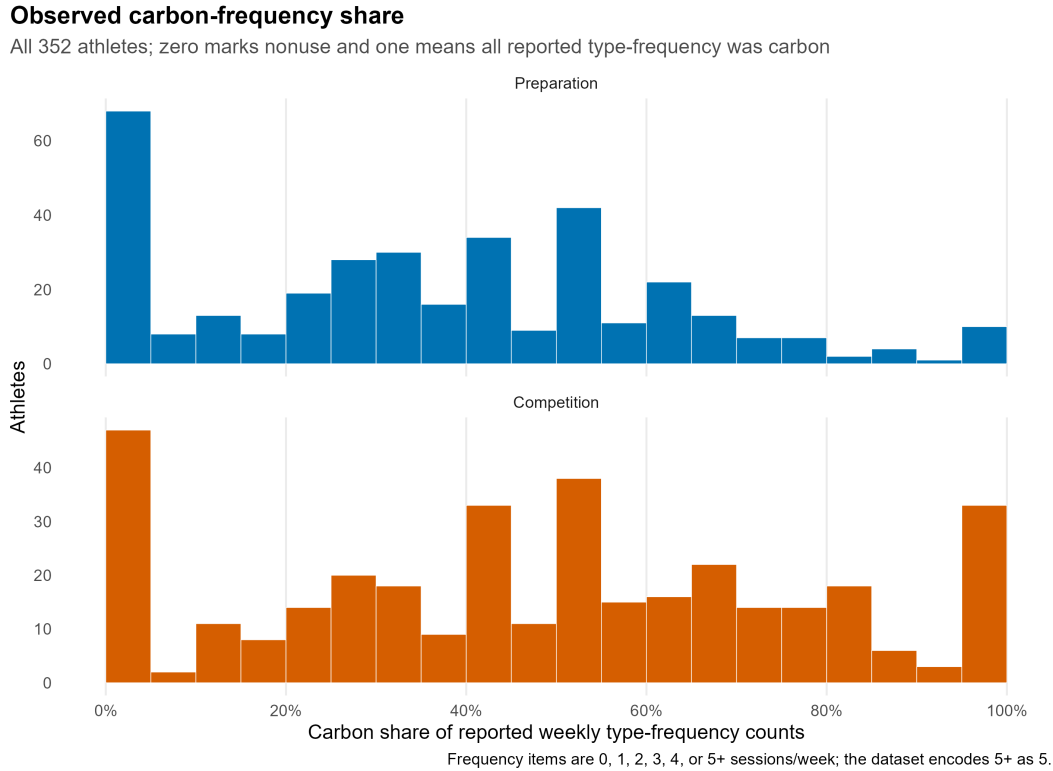


Figure 3: Observed distribution of carbon-plate insole share of reported training-type frequency, among all 352 athletes, by period.

Carbon-plate share of training is stable across sex and age group

Observed carbon share of reported training-type frequency, by period, age group, and sex

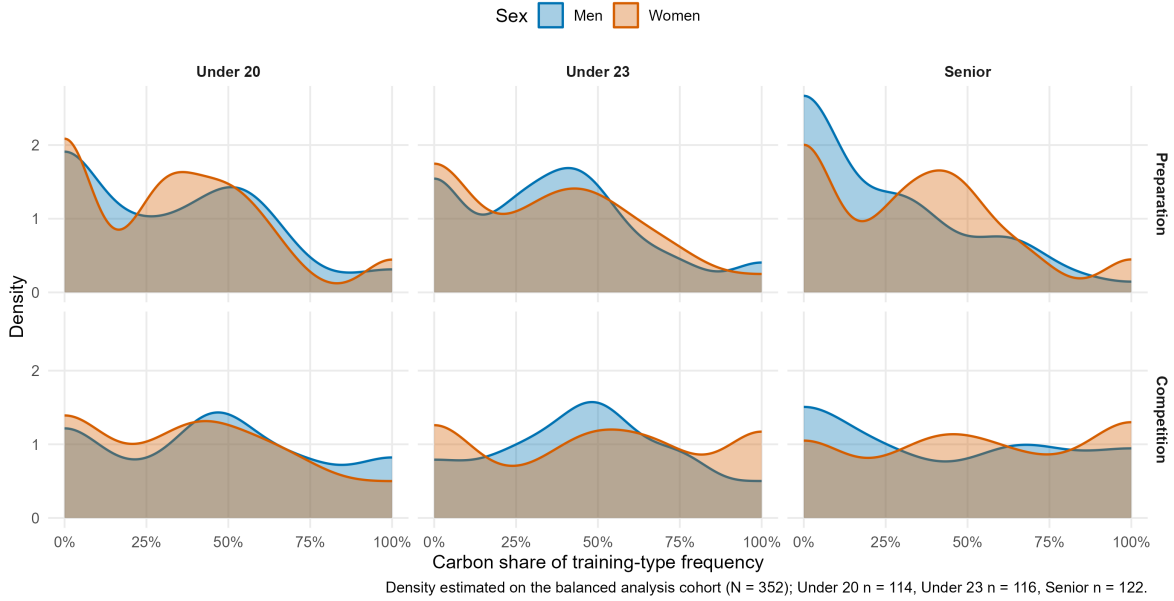


Figure 4: Observed distribution of carbon-plate insole share of reported training-type frequency, by period, age group, and sex. Curves are boundary-corrected kernel density estimates over the observed discrete-ratio share values (Section 2.5), shown to compare distributional overlap across strata; they are a smoothing device for visualization.

Posterior predictive distribution of injured-athlete counts

Replicated counts from the primary model (4000 posterior draws) versus the observed count, by period

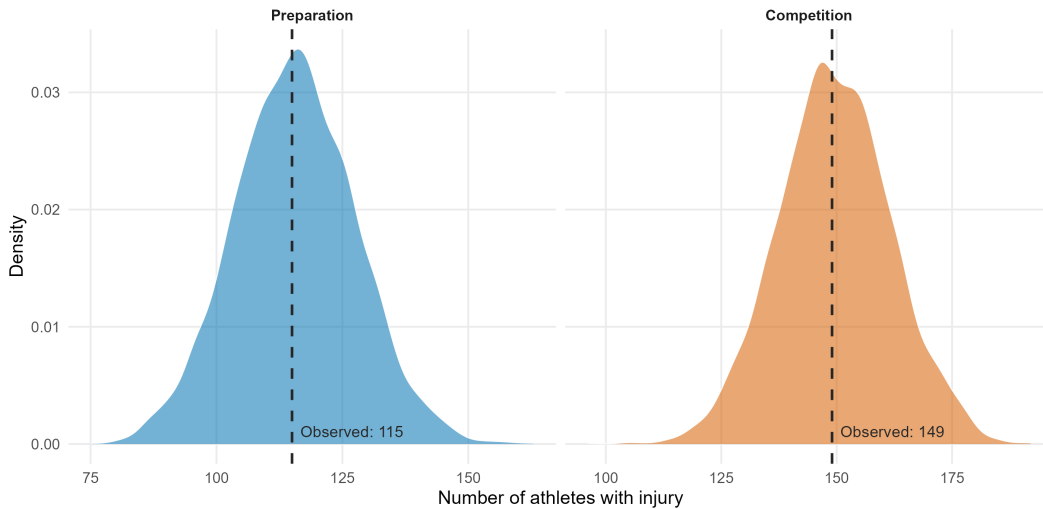


Figure 5: Posterior predictive check: full posterior distribution of the model-replicated number of injured athletes (4000 posterior draws of the primary model), by period. Dashed vertical line: observed count.

Full posterior distributions of adjusted associations

Adoption: median user vs nonuser; higher share: Q75 vs Q25 among users

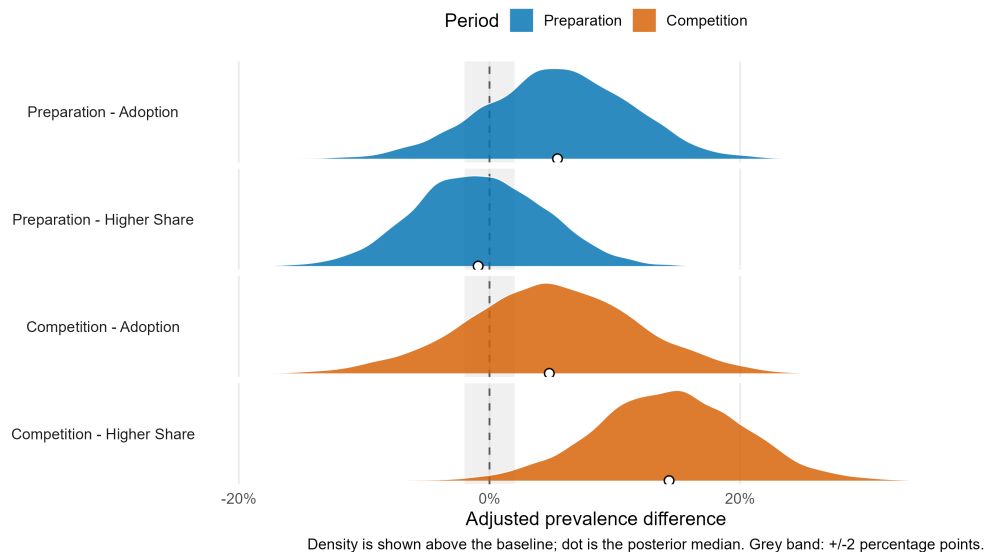


Figure 6: Full posterior distributions of the adjusted prevalence-difference estimand for the adoption and dose contrasts, by period. Grey band: region of practical equivalence (± 2 pp).

Fitted carbon-share dose response among users

Conditional injury odds relative to a user with 50% carbon share

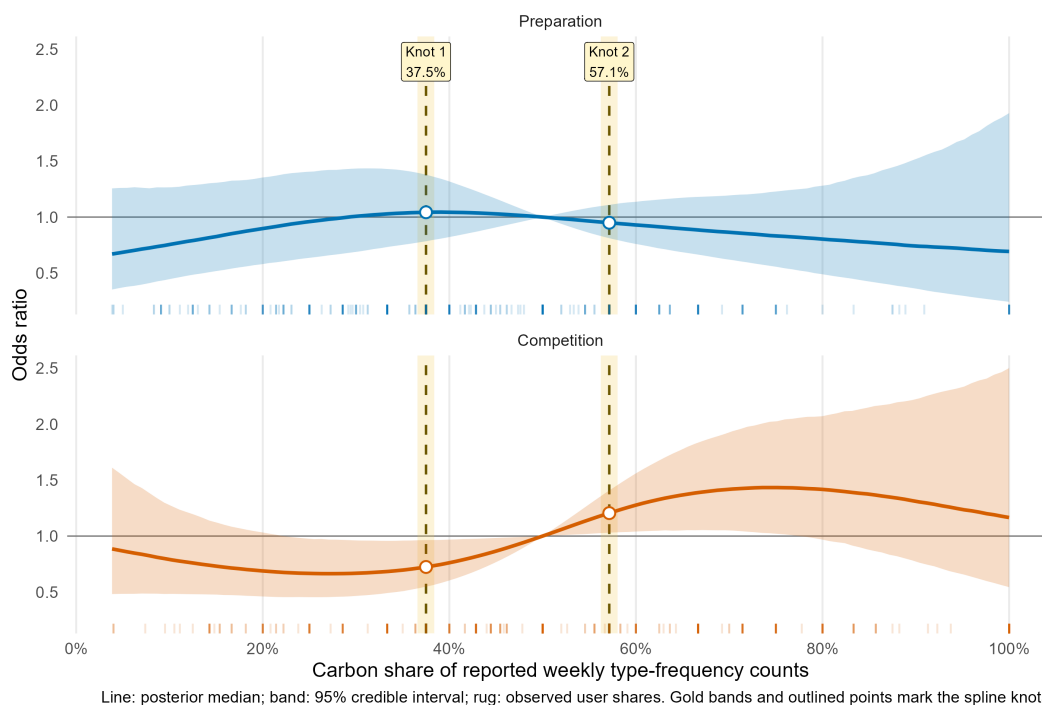


Figure 7: Fitted carbon-share dose response among users. Curves are conditional injury odds relative to a user with 50% carbon share; lines show posterior medians, bands 95% credible intervals, and rugs observed user shares. Gold bands, labeled vertical lines, and outlined points mark the internal spline knots (37.5% and 57.1%).

Full posterior distribution of the competition association

Q75 vs Q25 carbon share among competition-period users

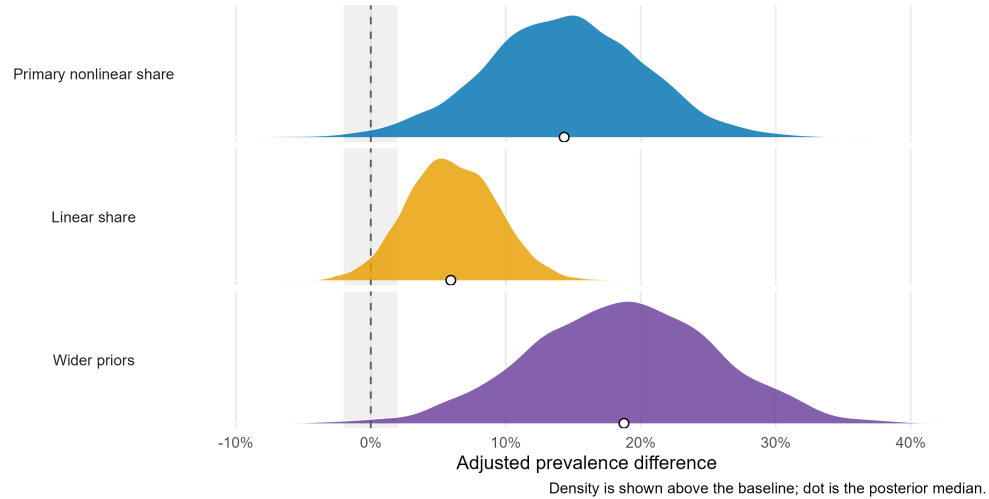


Figure 8: Sensitivity analysis: full posterior distribution of the competition-period dose association (75th vs. 25th percentile carbon share among users) under the primary nonlinear specification, a linear share specification, and wider priors.

Competition association conditional on sex

Target-standardized Q75 vs Q25 carbon share among users

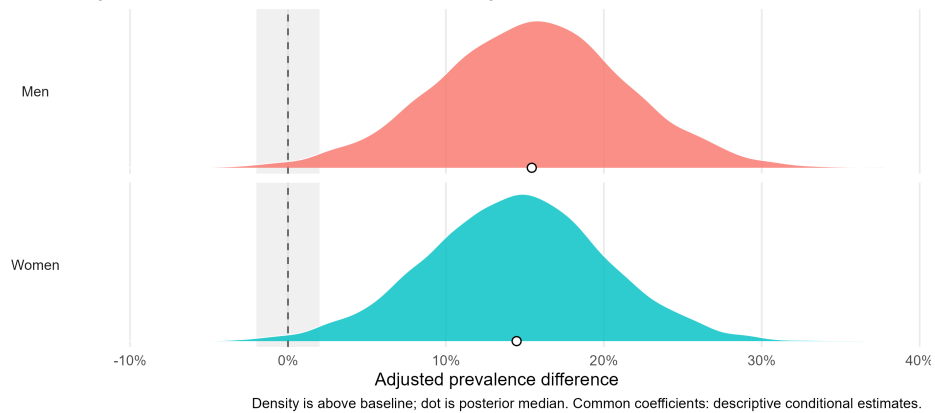


Figure 9: Competition-period dose association, population-weighted and stratified by sex.

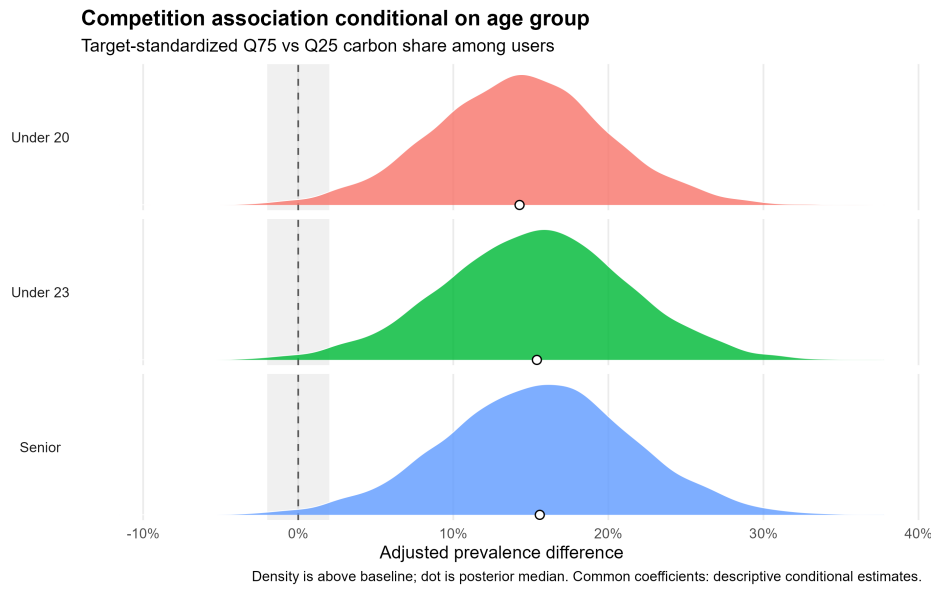


Figure 10: Competition-period dose association, population-weighted and stratified by age group.

Table 2: Baseline characteristics of the balanced analysis cohort ($N = 352$), by competition sex. Values are mean (SD), median [Q1, Q3], or n (%). SMD is the standardized mean difference, women vs. men (Cohen’s d for continuous/ordinal rows, computed on log-odds for proportions); $|\text{SMD}| \geq 0.20$ is conventionally a small imbalance and ≥ 0.50 a moderate one [Hayes-Larson et al., 2019]. Population-weighted columns apply the age-sex poststratification weights of Section 2.9 (Appendix Table 17).

Characteristic	Statistic	Men ($n = 194$)	Women ($n = 158$)	SMD sample	SMD weighted
Demographics					
Age, years	Mean (SD)	21.3 (3.1)	21.9 (3.1)	0.198	0.137
Age group: Under 20	n (%)	71 (36.6)	43 (27.2)	−0.202	−0.144
Age group: Under 23	n (%)	63 (32.5)	53 (33.5)	0.023	0.060
Age group: Senior	n (%)	60 (30.9)	62 (39.2)	0.175	0.093
Anthropometrics					
Height, cm	Mean (SD)	180.6 (5.9)	169.2 (6.8)	−1.809	−1.749
Weight, kg	Mean (SD)	70.5 (7.8)	57.8 (6.0)	−1.831	−1.847
Athletics history					
Experience, years	Mean (SD)	10.1 (4.4)	11.0 (3.9)	0.225	0.220
Injury					
Prior injury (previous 2 seasons)	n (%)	156 (80.4)	111 (70.3)	−0.237	−0.250
Any current-season injury	n (%)	120 (61.9)	84 (53.2)	−0.177	−0.210
Preparation-period injury	n (%)	69 (35.6)	46 (29.1)	−0.138	−0.165
Competition-period injury	n (%)	88 (45.4)	61 (38.6)	−0.137	−0.139
Carbon-plate insole exposure					
Any preparation carbon use	n (%)	159 (82.0)	128 (81.0)	−0.024	−0.041
Any competition carbon use	n (%)	173 (89.2)	135 (85.4)	−0.112	−0.130
Preparation carbon share (among users)	Median [Q1,Q3]	0.4 [0.2, 0.6]	0.4 [0.3, 0.6]	0.125	0.115
Competition carbon share (among users)	Median [Q1,Q3]	0.5 [0.3, 0.7]	0.5 [0.4, 0.8]	0.184	0.161
Training frequency (period total)					
Preparation total type-frequency	Median [Q1,Q3]	8 [6, 11]	8 [6, 13]	0.098	0.069
Competition total type-frequency	Median [Q1,Q3]	7 [5, 10]	7 [5, 11]	0.121	0.106
Discipline					
Sprint	n (%)	64 (33.0)	43 (27.2)	−0.126	−0.122
Middle distance	n (%)	38 (19.6)	31 (19.6)	0.001	0.015
Long distance	n (%)	36 (18.6)	23 (14.6)	−0.108	−0.064
Hurdles	n (%)	51 (26.3)	45 (28.5)	0.049	0.023
Horizontal jumps	n (%)	18 (9.3)	26 (16.5)	0.216	0.183
Pole vault	n (%)	10 (5.2)	9 (5.7)	0.024	0.001
Combined events	n (%)	16 (8.2)	25 (15.8)	0.234	0.195

Table 3: Sample flow from survey respondents to the balanced analysis cohort.

Stage	n
Consented, eligible, complete survey	355
Excluded: incomplete injury timing	-1
Excluded: zero total eligible frequency in a period	-2
Balanced analysis cohort (athletes)	352
Athlete-period records (preparation + competition)	704
Preparation-period injuries	115
Competition-period injuries	149

Table 4: Mean reported carbon share of training-type frequency, overall and by sex and age group. Carbon share is defined in Section 2.5 and includes non-users (share = 0).

Stratum	Level	n	Preparation, mean share	Competition, mean share
Overall	Overall	352	0.357	0.477
Sex	Men	194	0.349	0.468
	Women	158	0.366	0.488
Age group	Under 20	114	0.365	0.453
	Under 23	116	0.379	0.489
	Senior	122	0.328	0.488

Table 5: Characteristics and injury prevalence by specialty group. Groups collapse the seven athletics disciplines (Appendix Table 14) into four physiologically motivated categories: Sprint/hurdles (sprint or hurdles), Endurance (middle or long distance), Jumps (horizontal jumps or pole vault), and Combined events. Because disciplines and, consequently, groups are multi-select, 26 of 352 athletes belong to more than one group.

	<i>n</i>	Age, y	Women, %	Height, cm	Weight, kg	Exp., y	Prior inj., %	Any inj., %	Comp. inj., %
Sprint/hurdles	169	21.4 (3.0)	42.6	175.8 (8.7)	66.9 (9.3)	10.0 (3.9)	75.7	60.9	44.4
Endurance	109	21.6 (3.2)	42.2	174.0 (8.2)	59.6 (7.9)	10.3 (4.3)	67.9	44.0	30.3
Jumps	62	22.2 (3.5)	54.8	175.6 (9.0)	65.8 (9.6)	11.5 (4.6)	83.9	64.5	51.6
Combined events	41	22.0 (3.0)	61.0	176.6 (7.4)	67.8 (8.3)	11.9 (4.4)	82.9	70.7	48.8

Table 6: Carbon-plate insole use and training-type frequency by specialty group (grouping as in Table 5).

Specialty group	Prep. carbon share	Comp. carbon share	Prep. total freq., Mdn	Comp. total freq., Mdn
Sprint/hurdles	0.39	0.56	8	7
Endurance	0.31	0.34	10	9
Jumps	0.33	0.47	6	5
Combined events	0.41	0.52	9	9

Table 7: Anatomical location of reported injuries (259 injuries among 204 injured athletes; multi-select item, so location counts sum to more than 259).

Location	Injuries, n	% of injuries	% of injured athletes ($n = 204$)
Thigh (anterior or posterior)	97	37.5	39.2
Lower leg (anterior or posterior)	65	25.1	27.9
Foot (anterior or posterior)	50	19.3	22.1
Ankle (medial or lateral)	36	13.9	15.2
Groin	26	10.0	12.3
Hip	14	5.4	6.9
Head, upper limb, or trunk	11	4.2	5.4

Table 8: Type of reported lower-limb injuries. The type item was asked only for injuries localized to the hip, groin, thigh, lower leg, ankle, or foot (248 of 259 injuries; the questionnaire ended after the location item for the 11 head/upper-limb/trunk injuries); multi-select item, so counts sum to more than 248.

Type	Injuries, n	% of lower-limb injuries	% of injured athletes ($n = 204$)
Tendinopathy	71	28.6	29.9
Bone stress injury or stress fracture	65	26.2	25.0
Muscle strain	55	22.2	26.5
Other	39	15.7	16.7
Ligament injury	18	7.3	8.8
Plantar fasciopathy	16	6.5	7.8
Joint sprain	14	5.6	6.9
Tendon rupture	10	4.0	3.9

Table 9: Anatomical location of the three leading injury types (% of injuries of that type; multi-select, so rows sum to more than 100%). Location categories with 0% in all three rows (hip, groin, head/upper limb/trunk) are omitted.

Type (n injuries)	Thigh	Lower leg	Ankle	Foot
Tendinopathy (71)	22.5	33.8	22.5	35.2
Bone stress injury/stress fracture (65)	58.5	26.2	7.7	15.4
Muscle strain (55)	74.5	27.3	3.6	0.0

Table 10: Primary adjusted associations between carbon-plate insole exposure and injury prevalence, by period and contrast. Adjusted prevalence difference and ratio are posterior-standardized (Section 2.8.2); credible intervals are 95% equal-tailed. Adoption compares the median-share user with a non-user; dose compares the 75th with the 25th percentile of share among users.

Period	Contrast	Difference, pp (95% CrI)	Ratio (95% CrI)	P(positive)
Preparation	Adoption	5.4 (−6.3 to 16.2)	1.21 (0.84 to 1.74)	82.0%
Preparation	Dose	−0.8 (−10.6 to 9.1)	0.99 (0.73 to 1.30)	43.8%
Competition	Adoption	4.8 (−8.7 to 17.7)	1.16 (0.82 to 1.65)	77.1%
Competition	Dose	14.3 (2.7 to 25.8)	1.41 (1.06 to 1.86)	99.0%

Table 11: Sensitivity of the competition-period dose association (75th vs. 25th percentile share among users) to model specification.

Model	Difference, pp (95% CrI)	P(positive)
Primary (nonlinear spline share)	14.3 (2.7 to 25.8)	99.0%
Linear share	6.0 (−0.3 to 12.6)	96.9%
Wider priors	18.7 (5.2 to 31.9)	99.5%

4 Discussion

4.1 Key results

This national, two-phase-recruited sample of competitive Italian athletics athletes shows carbon-plated spikes established as the prevailing training equipment: a mean 35.7% of preparation-period and 47.7% of competition-period training-type frequency, with adoption above 80% in both periods and a consistent pattern across sex and age-group strata. After adjustment for demographic, anthropometric, training-load, and discipline covariates, adoption of carbon use was not clearly associated with injury prevalence in either period, and this held for preparation-period dose among users as well. Among competition-period users, a higher carbon share showed a credible association with higher adjusted injury prevalence: a posterior probability of 99.0% that the association was positive, a prevalence difference of 14.3 pp (95% CrI 2.7 to 25.8) between the 75th and 25th percentile of share, and a consistent direction across two structurally different sensitivity specifications and after population reweighting. The adapted OSTRC-H2 items added an athlete-centered burden estimate beyond injury counts: median severity was 75 of 100 and was highest among athletes reporting injuries in both periods. A further original descriptive finding reinforces this pattern: among the 248 lower-limb injuries with a recorded type, tendinopathy (28.6%) and bone stress injury or stress fracture (26.2%) were the two leading injury types, together exceeding muscle strain (22.2%)—a distribution markedly different from general athletics injury surveillance, where muscle injury typically predominates [Edouard et al., 2020]. Cross-tabulating type against location refines this further: tendinopathy was disproportionately concentrated at the foot, lower leg, and ankle relative to the injury sample as a whole, a distal pattern that lines up with the Achilles- and forefoot-tendon loading flagged by the biomechanical literature on carbon-plate use (Section 4.3); bone stress injury, by contrast, was thigh-dominant like the sample overall, so its elevated frequency in this cohort is a genuine finding but is not, on the present data, specifically localized to the foot or ankle.

4.2 Interpretation

The distinction between adoption and dose is central to how these results should be read, and it points to a coherent, biologically plausible pattern rather than a diffuse one. A uniform risk exposure would be expected to show an adoption association comparable to the dose association; instead, the credible association tracked specifically the amount of competition-period carbon use among athletes who had already adopted the technology. This is consistent with a threshold-like or cumulative-load mechanism, well supported by the biomechanical literature summarized in Section 1, in which greater longitudinal bending stiffness raises ankle joint contact force and plantarflexor peak force, particularly in athletes with lower baseline plantarflexor capacity [Gerlach et al., 2025, Willwacher et al., 2016]. The period specificity of the association—present in competition but not in preparation—fits naturally with competition-period training and racing involving greater absolute intensity and specificity of the loading that the carbon plate is designed to redistribute, a mechanistic hypothesis that this study’s design and results together motivate for direct testing.

The credible interval for the competition-period dose association excludes zero, and its posterior probability of exceeding a 5-pp practical-relevance threshold was 94.0%, a meaningful signal for a first population-level description of this exposure. The magnitude ranged from 6.0 to 18.7 pp across the two sensitivity specifications, indicating that the direction of the association is well established while its precise size remains an open, answerable question for future, larger studies. Taken together, the present evidence supports a clear, appropriately scoped conclusion: among competition-period carbon users in this cohort, a greater reported carbon share was associated with a greater adjusted

injury prevalence, after adjustment for the measured covariates. As a cross-sectional description, this finding is best understood as an association to be followed up prospectively, rather than as evidence of a causal effect of carbon-plate insoles on injury, of a risk from initiating carbon use, or of an equivalent pattern in the preparation period.

The OSTRC findings also show why injury burden should not be reduced to absence or complete cessation from sport. Time-loss-only systems can underestimate overuse problems that impair training, performance, or symptoms while athletes continue to participate [Clarsen et al., 2013, 2020]. Direct athlete reporting therefore complements injury counts and time loss with consequences that matter to the athlete. In this study the adapted score is best understood as a sport-specific athlete-reported outcome measure, not as a generic quality-of-life instrument or a substitute for prospectively repeated OSTRC surveillance.

4.3 Comparison with prior evidence

This study adds a population-level description to a literature that has so far relied on biomechanical and case-based work. A kinetic comparison of traditional and carbon-plated sprint spikes reported higher ankle joint contact force and peak plantarflexor force in the carbon-plated condition, with substantial inter-individual variability [Gerlach et al., 2025], providing a mechanistic account well aligned with the dose-sensitive pattern reported here. Case-based literature on Achilles tendon load with carbon-plate use [Tenforde et al., 2023] points in the same direction as the tendinopathy pattern reported here: tendinopathy was both over-represented as an injury type relative to the typical athletics pattern [Edouard et al., 2020] and disproportionately located at the foot, lower leg, and ankle (Table 9), consistent with that case-based signal at the population level for the first time. The same case-based literature has also flagged navicular and cuneiform bone stress with carbon-plate use [Tenforde et al., 2023]; this cohort’s elevated overall rate of bone stress injury is compatible with that concern in aggregate, but the injuries were thigh-dominant rather than foot-dominant in location, and the survey did not resolve bone-specific site (for example, navicular versus tibial), so this cohort’s data support the tendinopathy hypothesis more directly than the site-specific bone-stress-injury hypothesis. To the authors’ knowledge, this is among the first studies to describe carbon-plate insole exposure and its association with injury prevalence at the population level in a national sample, extending this prior mechanistic and case-based work with cohort-level evidence; the tendinopathy and bone-stress-injury patterns observed here are hypothesis-generating findings that a dedicated, adequately powered type-stratified analysis should test directly.

4.4 Strengths and considerations for interpretation

The two-phase recruitment, national age–sex poststratification, and Bayesian multilevel design were built specifically to support a defensible population-level estimate from survey data, and several features of the results reflect that: consistent direction across sensitivity specifications, stable estimates after reweighting, well-behaved sampling diagnostics, and posterior predictive checks that closely reproduced observed injury counts. A number of features of the data are useful to keep in mind when applying these results. Because exposure and outcome were recalled together for the same season, this cross-sectional design describes an association rather than establishing temporal order between carbon-share increase and injury onset; a prospective design with objectively measured exposure is the natural next step and one this study’s findings directly motivate. Carbon share was derived from ordinal weekly training-type-frequency items, a practical and scalable measure for a national survey, though a more granular, objectively measured training-time or footwear-wear-duration metric would sharpen the dose estimate further. The competition-period non-user group, while smaller

(44 of 352 athletes) than the user group, reflects the near-universal adoption of carbon-plate spikes in this population and is itself a noteworthy descriptive finding; it does widen the credible interval for the adoption contrast. The dose-association magnitude varied with the assumed functional form and prior width, so the 14.3-pp estimate is best read as indicating a positive association of plausibly meaningful size rather than a fixed effect size—precisely the kind of estimate that a focused follow-up study, informed by this result, is well placed to refine. The adjustment set covered the principal risk factors identified in prior running-injury literature [Taunton et al., 2003]; additional factors such as coaching-driven equipment allocation or baseline plantarflexor strength were not measured here and represent promising directions for future covariate collection. Self-report of both exposure and outcome is standard practice in large-scale sports injury surveillance [Bahr et al.] and enabled a national sample of this scale; validating the survey items against objective training logs is a natural complement for future work. The adapted OSTRC-H2 items were completed by 94.6% of injured athletes but were asked only once and only after injury was reported, so they cannot recover weekly prevalence, symptom duration, cumulative burden, or scores among athletes who did not classify themselves as injured; the period-stratified SMDs are correspondingly descriptive. The study protocol was prospectively reviewed and approved by the Comitato Etico of the University of Turin (Section 2.2), providing a documented, ethics-verified record of pre-specification alongside this manuscript.

4.5 Generalizability

The two-phase recruitment and age–sex poststratification were designed specifically to support generalizability to the national competitive population; the modest Kish effective-sample-size retention after weighting (94%) and the close agreement between weighted and unweighted estimates (Appendix Table 19) support good generalizability to Italian FIDAL-affiliated competitive athletes across the age categories studied. Extending these findings to recreational athletes, road-running populations, or national contexts with different equipment-access patterns is a promising direction for future work, particularly given the near-universal adoption of carbon-plate spikes observed in this competitive cohort.

4.6 Conclusion

In this national cohort of competitive Italian athletics athletes, carbon-plated spikes were the prevailing training equipment in both preparation and competition, with a stable, roughly two-fifths-to-one-half share of reported training-type frequency across sex and age-group strata. Adoption of carbon-plate use was not clearly associated with injury prevalence. Among competition-period users, a greater reported carbon share showed a credible, dose-sensitive association with higher adjusted injury prevalence, consistent in direction across sensitivity analyses and population reweighting. These findings offer a clear, actionable basis for prospectively designed surveillance of carbon-plate insole exposure and injury in athletics, using objectively measured exposure and known temporal order, and support engaging athletes, coaches, and clinicians in monitoring competition-period carbon-plate dose as this evidence base develops.

Ethics

This study was approved by the Comitato Etico of the University of Turin (protocol n. 0453001, approved 23 June 2026). All athletes provided electronic informed consent before completing the

survey; athletes under the age of majority provided assent with parental/guardian consent as required by the approved protocol. Data were pseudonymized at the point of collection and analyzed under a study identifier; no directly identifying information (name, contact details, club affiliation) is reported in this manuscript or in any derived table or figure, in line with the data-privacy provisions of the Declaration of Helsinki and with STROBE-SIIS item SIIS 23.1 [Bahr et al.]. The two-phase recruitment and poststratification procedures described in Section 2.4 and Section 2.9 were part of the ethics-approved protocol.

Funding

None declared.

Data and code availability

Analysis code, aggregate tables, and the reproducible manuscript source are publicly available at <https://github.com/MVittu/ATLETA>. Individual-level survey responses and detailed recruitment-monitoring cells are not publicly shared because participant consent and ethics safeguards do not authorize open release.

References

- Roald Bahr, Benjamin Clarsen, Wayne Derman, Jiri Dvorak, Carolyn A. Emery, Caroline F. Finch, Martin Hagglund, Astrid Junge, Simon Kemp, Karim M. Khan, Stephen W. Marshall, Willem Meeuwisse, Margo Mountjoy, John W. Orchard, Babette Pluim, Ken L. Quarrie, Bruce Reider, Martin Schwellnus, Torbjorn Soligard, Keith A. Stokes, Toomas Timpka, Evert Verhagen, Abhinav Bindra, Richard Budgett, Lars Engebretsen, Ugur Erdener, and Karim Chamari. International olympic committee consensus statement: Methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including the STROBE extension for sports injury and illness surveillance (STROBE-SIIS)).
- Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017. doi: 10.18637/jss.v076.i01.
- Benjamin Clarsen, Grethe Myklebust, and Roald Bahr. Development and validation of a new method for the registration of overuse injuries in sports injury epidemiology: The Oslo Sports Trauma Research Centre (OSTRC) overuse injury questionnaire. *British Journal of Sports Medicine*, 47(8):495–502, 2013. doi: 10.1136/bjsports-2012-091524.
- Benjamin Clarsen, Ola Rønsen, Grethe Myklebust, Tonje Wåle Flørenes, and Roald Bahr. The Oslo Sports Trauma Research Center questionnaire on health problems: A new approach to prospective monitoring of illness and injury in elite athletes. *British Journal of Sports Medicine*, 48(9):754–760, 2014. doi: 10.1136/bjsports-2012-092087.
- Benjamin Clarsen, Roald Bahr, Grethe Myklebust, Stig Haugsbø Andersson, Sean Iain Docking, Michael Drew, Caroline F. Finch, Lauren Victoria Fortington, Joar Harøy, Karim M. Khan, Bill Moreau, Isabel S. Moore, Merete Møller, Dustin Nabhan, Rasmus Oestergaard Nielsen, Kati Pasanen, Martin Schwellnus, Torbjørn Soligard, and Evert Verhagen. Improved reporting of

- overuse injuries and health problems in sport: An update of the Oslo Sport Trauma Research Center questionnaires. *British Journal of Sports Medicine*, 54(7):390–396, 2020. doi: 10.1136/bjsports-2019-101337.
- Pascal Edouard, Laurent Navarro, Pedro Branco, Vincent Gremeaux, Toomas Timpka, and Astrid Junge. Injury frequency and characteristics (location, type, cause and severity) differed significantly among athletics (‘track and field’) disciplines during 14 international championships (2007–2018): Implications for medical service planning. *British Journal of Sports Medicine*, 54(3):159–167, 2020. doi: 10.1136/bjsports-2019-100717.
- Andrew Gelman. Struggles with survey weighting and regression modeling. *Statistical Science*, 22(2):153–164, 2007. doi: 10.1214/088342306000000691.
- Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, Boca Raton, FL, 3rd edition, 2013.
- Marius Gerlach, Marie-Louise de Graaf, Kim J. Boström, and Heiko Wagner. Effect of advanced footwear technology spikes on lower limb kinetics and sprint performance. *Footwear Science*, 17(2):69–77, 2025. doi: 10.1080/19424280.2025.2458322.
- Paul A. Harris, Robert Taylor, Robert Thielke, Jonathon Payne, Nathaniel Gonzalez, and Jose G. Conde. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *Journal of Biomedical Informatics*, 42(2):377–381, 2009. doi: 10.1016/j.jbi.2008.08.010.
- Paul A. Harris, Robert Taylor, Brenda L. Minor, Veida Elliott, Michelle Fernandez, Lindsay O’Neal, Laura McLeod, Giovanni Delacqua, Francesco Delacqua, Jacqueline Kirby, and Stephany N. Duda. The REDCap consortium: Building an international community of software platform partners. *Journal of Biomedical Informatics*, 95:103208, 2019. doi: 10.1016/j.jbi.2019.103208.
- Eleanor Hayes-Larson, Katrina L. Kezios, Stephen J. Mooney, and Gina Lovasi. Who is in this study, anyway? guidelines for a useful Table 1. *Journal of Clinical Epidemiology*, 114:125–132, 2019. doi: 10.1016/j.jclinepi.2019.06.011.
- Laura A. Healey, Mathieu Bertschy, Shalaya Kipp, and Wouter Hoogkamer. Can we quantify the benefits of “super spikes” in track running? *Sports Medicine*, 52(6):1211–1218, 2022. doi: 10.1007/s40279-022-01657-4.
- Daniel Martínez-Silván, Eirik H. Wik, Juan M. Alonso, Elodie Jeanguyot, Bojana Salcinovic, and Andrew Johnson. Injury characteristics in male youth athletics: A five-season prospective study in a full-time sports academy. *British Journal of Sports Medicine*, 55(17):954–960, 2021. doi: 10.1136/bjsports-2020-102373.
- Joel Mason, Dominik Niedziela, Jean-Benoît Morin, Andreas Groll, and Astrid Zech. The potential impact of advanced footwear technology on the recent evolution of elite sprint performances. *PeerJ*, 11:e16433, 2023. doi: 10.7717/peerj.16433.
- Alannah K. A. McKay, Trent Stellingwerff, Ella S. Smith, David T. Martin, Iñigo Mujika, Victoria L. Goosey-Tolfrey, Jeremy Sheppard, and Louise M. Burke. Defining training and performance caliber: A participant classification framework. *International Journal of Sports Physiology and Performance*, 17(2):317–331, 2022. doi: 10.1123/ijsp.2021-0451.

- John M. Neuhaus, John D. Kalbfleisch, and Walter W. Hauck. A comparison of cluster-specific and population-averaged approaches for analyzing correlated binary data. *International Statistical Review*, 59(1):25–35, 1991. doi: 10.2307/1403572.
- Jack E. Taunton, Michael B. Ryan, Douglas B. Clement, Donald C. McKenzie, David R. Lloyd-Smith, and Bruno D. Zumbo. A prospective study of running injuries: The Vancouver Sun Run “in training” clinics. *British Journal of Sports Medicine*, 37(3):239–244, 2003. doi: 10.1136/bjsm.37.3.239.
- Adam S. Tenforde, Tim Hoenig, Amol Saxena, and Karsten Hollander. Bone stress injuries in runners using carbon fiber plate footwear. *Sports Medicine*, 53(8):1499–1505, 2023. doi: 10.1007/s40279-023-01818-z.
- Toomas Timpka, Juan-Manuel Alonso, Jenny Jacobsson, Astrid Junge, Pedro Branco, Benjamin Clarsen, Jan Kowalski, Margo Mountjoy, Sverker Nilsson, Babette Pluim, Per Renström, Ola Ronsén, Kathrin Steffen, and Pascal Edouard. Injury and illness definitions and data collection procedures for use in epidemiological studies in athletics (track and field): Consensus statement. *British Journal of Sports Medicine*, 48(7):483–490, 2014. doi: 10.1136/bjsports-2013-093241.
- Toomas Timpka, Jenny Jacobsson, Joakim Ekberg, Caroline F. Finch, Jerome Bichenbach, Pascal Edouard, Michael F. Bergeron, Willie O. Roberts, Juan-Manuel Alonso, and Per A. Renström. Meta-narrative analysis of sports injury reporting practices based on the injury definitions concept framework (IDCF): A review of consensus statements and epidemiological studies in athletics (track and field). *Journal of Science and Medicine in Sport*, 18(6):643–650, 2015. doi: 10.1016/j.jsams.2014.11.393.
- Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432, 2017. doi: 10.1007/s11222-016-9696-4.
- Erik von Elm, Douglas G. Altman, Matthias Egger, Stuart J. Pocock, Peter C. Gøtzsche, and Jan P. Vandenbroucke. The strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *The Lancet*, 370(9596):1453–1457, 2007. doi: 10.1016/S0140-6736(07)61602-X.
- Steffen Willwacher, Manuel Kurz, Christian Menne, Elmar Schrödter, and Gert-Peter Brüggemann. Biomechanical response to altered footwear longitudinal bending stiffness in the early acceleration phase of sprinting. *Footwear Science*, 8(2):99–108, 2016. doi: 10.1080/19424280.2016.1144653.

A STROBE-SIIS reporting checklist

Table 12 follows the STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) checklist [Bahr et al.], itself an extension of the STROBE statement for observational studies [von Elm et al., 2007]. The final column gives the manuscript page reporting each item, resolved automatically at compilation so that it remains accurate under re-typesetting; line numbers are printed in the margin throughout the manuscript (via `lineno`) to allow more precise location within the cited page.

Table 12: STROBE-SIIS reporting checklist, with the manuscript page reporting each item.

Item	Recommendation	Section	Page
1a	Study design term in title/abstract	Title	p. 1
1b, SIIS1.1, SIIS1.2	Informative abstract; sport/population/level; duration of observation	Abstract	p. 1
2	Background and rationale	Introduction	2
3, SIIS3.2	Objectives and specific purpose	Introduction	2
SIIS3.1	Registration status	Study registration	3
4, SIIS4.4	Study design; athlete/sport categorization	Design; Covariates	3, 5
SIIS4.1	Health problems observed	Outcome	5
SIIS4.2, SIIS4.3	Recording approach and coding	Exposure; Outcome	4, 5
5, SIIS5.1–5.3	Setting, location, dates, prospective/retrospective	Setting; Participants	3, 4
6	Eligibility, selection, recruitment	Participants	4
7, SIIS7.1, SIIS7.2	Variable definitions; outcome identification	Exposure; Outcome	4, 5
8, SIIS8.1–SIIS8.5	Data sources and measurement; timing; duration	Setting; Bias	3, 8
9, SIIS9.1, SIIS9.2	Bias and validation/reliability	Bias	8
10	Study size derivation	Participants	4
11, SIIS11.1, SIIS11.2	Quantitative variables; multiple-problem handling; severity	Outcome; Exposure	5, 4
12a	Statistical methods	Statistics	5
12b	Subgroup/interaction methods	Subgroups	8
12c	Missing-data handling	Missing data	9
12d	Sampling-strategy analytic accounting	Weighting	8
12e	Sensitivity analyses	Validation	8
SIIS12.1	Exposure-to-risk adjustment	Statistics	5
SIIS12.2	Risk-measure calculation	Estimands	6
SIIS12.4	Multiple health-problem handling	Outcome	5
SIIS12.5	Handling of athletes not included at out-set	Missing data	9
SIIS12.6	Follow-up/censoring management	Weighting	8
13, SIIS13.1–SIIS13.3	Participant flow and reasons for exclusion	Sample flow	9
14, SIIS14.1	Descriptive characteristics by level of competition	Baseline characteristics	9
15, SIIS15.1	Outcome data; athletes vs. problems distinction	Descriptive use; Primary results	10, 11
16, SIIS16.1, SIIS16.2	Main adjusted results with precision	Primary results	11
17, SIIS17.1	Other analyses; injury region/type in tabular form	Injury pattern; Sensitivity; Weighted/subgroup	10, 13, 13
18	Key results summary	Discussion	23
19, SIIS19.1	Limitations	Limitations	24
20	Interpretation	Interpretation	23
21, SIIS21.1	Generalizability	Generalizability	25
22	Funding	Funding	26
23, SIIS23.1	Ethics and data privacy	Ethics	25

B Software and session information

All statistical analyses were conducted in R. Table 13 lists the software, model family, and sampling configuration used for the primary Bayesian multilevel model and its validation; runs used base seed 20260719 with deterministic model-specific offsets for reproducibility.

Table 13: Software, model, and posterior-sampling configuration.

Component	Version
R	4.5.2 (2025-10-31 ucrt), R Core Team
Platform toolchain	Rtools 4.5 (x86_64-w64-mingw32)
rstan	2.32.7
StanHeaders	2.32.10
loo	2.8.0
splines	4.5.2 (base R)
Model family	Bernoulli likelihood with logit link
Sampler	Stan Hamiltonian Monte Carlo, No-U-Turn Sampler
Chains $\times$ iterations	4 $\times$ 2000 per fitted model
Warm-up/adaptation	1000 iterations per chain (discarded)
Retained posterior draws	1000 per chain; 4000 per fitted model
Target acceptance probability	<code>adapt_delta</code> =0.98
Maximum tree depth	12
Random seed	20260719 plus deterministic model-specific offsets

Model code (`model.stan`, `weighted_model.stan`), the full analysis pipeline (`analysis.R`, `population_analysis.R`), and every derived table and figure referenced in this manuscript are retained under `experiments/multilevel/` in the project repository. Re-running `experiments/multilevel/analysis.R` reproduces Tables 3, 10, 11, and 20 and Figures 3–6 from the source data file referenced in Section 2.4.

C Supplementary tables

Table 14: Discipline support in the balanced analysis cohort. Codes 8–11 (relay, throws, race walk, and one further unused code) had zero respondents and were excluded from discipline-level adjustment.

Code	Discipline	n	Included in model
1	Sprint	107	Yes
2	Middle distance	69	Yes
3	Long distance	59	Yes
4	Hurdles	96	Yes
5	Horizontal jumps	44	Yes
6	Pole vault	19	Yes
7	Combined events	41	Yes
8–11	(zero respondents)	0	No

Table 15: FIDAL competition-registry counts used to construct the 2026 eligible-population frame. Under 20 counts for 2026 were used directly. Because 2026 counts were not yet available for Under 23 and Senior athletes when recruitment was planned, their sex-specific frame estimates were the rounded means of the 2023–2025 counts. The resulting age–sex frame totaled 1328 athletes; 2025 discipline shares within each age–sex cell were used to allocate stratum targets.

Age group	Sex	2023	2024	2025	2026	Frame estimate
Under 20	Women	235	265	274	204	204
Under 20	Men	327	361	338	311	311
Under 23	Women	217	211	207	–	212
Under 23	Men	244	247	252	–	248
Senior	Women	176	176	156	–	169
Senior	Men	195	199	159	–	184
Total		1394	1459	1386	–	1328

Table 16: Bayesian worst-case population-proportion precision calculation used to set Phase 2 (targeted recall) enrollment targets. The calculation set a minimum recruitment target; representativeness was addressed separately through stratified recall and poststratification.

Quantity	Value
Planning quantity	Population proportion
Prior	Beta(1,1)
Worst-case count	$x = \lfloor n/2 \rfloor$
Posterior at candidate n	Beta($1 + x, 1 + n - x$)
Interval and target	Equal-tailed 95% CrI; half-width ≤ 0.06
$n = 263$ half-width	0.060038
$n = 264$ half-width	0.059926
Estimated eligible population	1328
Minimum target sample	264 (19.9% of frame)
Consented, eligible, complete	355 (26.7% of frame)
Balanced analysis cohort	352 (26.5% of frame)
Target allocation	Proportional across 42 strata

Table 17: Age–sex poststratification weight diagnostics (Section 2.9). Model weight is the regularized ratio of target population share to sample share; Kish effective sample size after weighting was 340.5 (97% of $N = 352$).

Age group	Sex	Target n	Sample n	Sample share	Model weight
Under 20	Women	204	43	12.2%	1.26
Under 23	Women	212	53	15.1%	1.06
Senior	Women	169	62	17.6%	0.72
Under 20	Men	311	71	20.2%	1.16
Under 23	Men	248	63	17.9%	1.04
Senior	Men	184	60	17.0%	0.81

Table 18: Age- and sex-conditional posterior estimate of the competition-period dose association (75th vs. 25th percentile carbon share among users), holding the primary model structure and covariate adjustment fixed. These are descriptive, hypothesis-generating comparisons (Section 2.10), not independently powered tests of effect modification.

Grouping	Level	Difference, pp (95% CrI)	P(positive)
Sex	Men	15.4 (3.3 to 26.9)	99.4%
Sex	Women	14.4 (3.2 to 25.5)	99.4%
Age group	Under 20	14.3 (3.1 to 25.3)	99.4%
Age group	Under 23	15.3 (3.3 to 26.9)	99.4%
Age group	Senior	15.5 (3.4 to 27.1)	99.4%

Table 19: Population-weighted versus sample (unweighted) adjusted estimands. Weighting applies the age-sex poststratification of Table 17.

Period	Contrast	Unweighted, pp (95% CrI)	Weighted, pp (95% CrI)
Preparation	Adoption difference	5.4 (−6.3 to 16.2)	4.3 (−6.5 to 14.7)
Preparation	Dose difference	−0.8 (−10.6 to 9.1)	−0.3 (−10.5 to 9.6)
Competition	Adoption difference	4.8 (−8.7 to 17.7)	7.3 (−5.6 to 19.8)
Competition	Dose difference	14.3 (2.7 to 25.8)	14.9 (3.3 to 26.4)

Table 20: Model comparison and validation diagnostics for the primary (exposure) and matched baseline (no-exposure) models. ELPD: expected log predictive density (athlete-grouped PSIS-LOO); all Pareto- $k \leq 0.5$ for both models.

Model	ELPD _{LOO}	SE	Max $\hat{R}$	Divergences
Primary (exposure)	−428.48	11.13	1.003	0
Baseline (no exposure)	−430.70	10.98	1.002	0
ELPD difference (exposure − baseline): 2.22 (SE 2.30)				